

Efficiency-Triggered Self-Powered Solar Panel Cleaning: Threshold-Based Dust Mitigation and Scalability Analysis for IoT-Enabled PV Systems

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Abstract—Accumulated dust on solar panels severely impacts their efficiency and results in considerable energy loss, particularly in outdoor locations that are extremely dusty. Conventional cleaning techniques are generally done manually, are inefficient, and are not designed for use in ongoing maintenance. This paper discusses and presents a fully autonomous solar panel cleaning system that includes real-time performance monitoring through utilization of an ESP32 microcontroller and INA219 sensor. This system continuously measures panel output and environmental factors and only activates cleaning operations when it has been determined that the efficiency of the solar panels is significantly below optimum levels for the ambient level of sunlight that is received. In order to deliver the required cleaning, a motor-operated mechanism is integrated with a pump to provide water flow across the solar panel, therefore effectively removing surface contaminants. Laboratory and field test results confirm that dust accumulation can devalue the output of a solar panel by as much as 40%. Using this system, the output of the solar panels would be restored by up to 90-92% of the original performance, thereby resulting in an overall net positive energy balance because the total energy consumption of the entire system is less than the amount of energy that would have been lost through the accumulation of dust. This system provides an efficient, scalable, and cost-effective solution for maintaining photovoltaic generation efficiency.

Keywords—Photovoltaic cells (PV), automated cleaning mechanism, dust mitigation, energy efficiency, wiper cleaning, water conservation, IoT, solar panel.

I. INTRODUCTION

Solar technology is one of the most rapidly expanding energy sources in the world, as evidenced by the dramatic rise in combined installed capacity of PV systems globally, exceeding 1.5TW in 2023 and nearly 74% of all newly installed renewable electricity generating capacity. The International Energy Agency projects that under net-zero emission scenarios, solar will be the leading source of electricity generation by 2050 [1]. While there are many advantages to using photovoltaics, one of the most significant challenges to their operation is degradation in performance due to the accumulation of dust on PV modules. The National Renewable Energy Laboratory has found that when not cleaned on a regular basis, soiling can reduce the energy

output of solar panels by approximately 2 - 7% under normal conditions and up to 20 - 40% in dry, arid regions, and potentially over 50% in prolonged unmaintained, highly polluted deserts [2]. Studies show that power losses due to dust can range from 15% to over 50% depending on different environmental conditions [3]. Because of the narrow operating margins of solar projects today, even a 5% decline in efficiency over a 12-month period could significantly reduce the annual energy production and economic return. Old-fashioned methods to clean solar panels are mostly done by hand, taking a long time and require extra water, so cleaning panels that are on rooftops is inefficient as well. Automated cleaning systems that do exist tend to be expensive and/or have a complicated structure and don't provide real-time performance data. The current research proposes to overcome these challenges with the development of an affordable, programmable cleaning system for solar panels integrating real-time monitoring of photovoltaic power generation. This system is built on condition rather than scheduled cleaning i.e., only cleaning when there is a large enough decrease in energy generation.

II. LITERATURE REVIEW

Moungsrijun et al. [4] in the article describes the creation and deployment of a cleaning robot for PV panels that utilises control through an Arduino microprocessor. Dust build-up on the surface of PV panels can lead to significant power loss of up to 50% in dirty environments. The cleaning robot consists of DC motors for propulsion, plastic brushes or soft foam sponges for cleaning, a microprocessor to control cleaning track movement, and a camera to assist in locating obstacles and mapping the PV panel area.

Purnata et al. [5] describes a mobile solar panel cleaning robot with an improved Kalman filter algorithm used to determine the exact location of the robot using Inertial Measurement Units (IMU) sensor data derived from the z axis for accurate navigation in 3D coordinates. The robot operates as a dry cleaning unit using sweeping mechanisms, along with cooling flows to help dissipate dust and lower the temperature of solar panels by 5 to 6 degrees Celsius (°C), thus addressing issues related to dust accumulation and overheating due to heat loss.

The robotic system outlined in the study by Murugaperumal et al., [6] focuses on the automated removal of dust from solar PV panels. This solution addresses a significant issue of up to

40% efficiency loss due to dust build-up by using GPS navigation systems to replace much of the human intervention required.

The article by **Akyazi et al.**, [7] presents the design and real time testing of a robotic cleaning system for solar panels to attain higher efficiency rates. It is a dual-motor crawler robot that moves horizontally and has a cleaning brush which operates vertically, even using position switches for area control. Noh et al., [8] presents the holistic development of a fully autonomous solar panel cleaning robot that works using an Arduino UNO. The robot works using ultrasonic sensing, dual brushes and a water pump with the goal of removing dust from solar panels. Experimental results where it was tested show that the electrical output increases by 50% in dusty situations, note that this was carried out using flour to act as dust.

Kimber et al [9] in this paper measure the effects of soiling on large PV systems that are connected to the grid in California and the southwestern United States, where dryness leads to greater dirt accumulation on PV panels. Using a linear degradation model of performance with rainless days, the study estimated degradation rates between 0.1 to 0.3% per day depending on location, with annual losses due to the effects of soiling of between 2–6%. The paper by Uçkan et al., [10] presents a cost effective automated solar panel cleaning system controlled by Raspberry Pi, aimed at reducing dust accumulation on solar panels and eventually increasing efficiency. The system detects dust levels and uses an autonomous mechanical cleaning system to remove the debris, typically using brushes. The experimental results emphasized that the system maintains low power consumption.

Elbrashy et al., [11] provides a comprehensive review of robotic cleaning technologies that clean PV panels to address the critical issue of the accumulation of dust that reduces panel efficiency up to 40% in certain conditions. In this paper robots are highlighted as efficient in cleaning up to 99% of the surface contaminants, that too without water usage. The paper emphasises on the integration of sensors and AI based integration for the cleaning of panels.

Ghodki et al., [12] in this paper present a novel IR and sprinkler based embedded controller robotic arm for the automatic cleaning of solar panels to mitigate dust accumulation, significantly reducing photovoltaic efficiency. The system uses IR sensors to detect dust as well as controlling thermal conditions. This robot uses a silicon rubber wiper and PWM controlled motors to clean in wet conditions successfully. Experimental results with a 30 WP prototype showed an 11.26% increase in energy output and a 13.63% improvement in module efficiency, as a whole there was a 85.20% performance ratio after a 73 day testing period which proved the systems efficiency.

The paper by **Mote et al.**, [13] presents an Arduino based solar panel cleaning system made to reduce efficiency losses by dust accumulation on photovoltaic (PV) panels. This system works using mechanical brushing and controlled water spraying to eliminate the dust and debris with as minimal human intervention as possible. Field tests demonstrated that there was a 25-40% increase in energy output with minimal water usage of only 200 mL per cycle. The system costs under 5000 rupees and is scalable for large scale installations.

The paper by **N. Ronnaronglit et al.**, [14] describes the design and development of an autonomous robotic system to clean solar panels to increase the energy output. This system aimed to develop such robots suitable for the climate of Thailand. It used a wireless joystick, sensor using a gear motor and an arduino microcontroller to control the system. The cleaning of the actual panel is done with a rotary brush and a water spray which can operate at surface level at a 0-30 degree Celsius. Experimental results show that it would clean up to 80% of the panels.

This paper by **Chakaborty et al.** [15] presented a smart remote solar panel cleaning robot which could communicate wirelessly. To overcome energy efficiency reduction a robot was built which can be controlled using mobile phones that can be operated remotely through bluetooth. The system proves to be efficient through experimental results and becomes most useful in large scale solar panel installations where energy losses pile up.

The robotic system outlined in the paper by **Suleiman et al.** [16] proposes a solution to the increasing need for maintenance of solar panels with the aim of maximising energy output. The study presents the design and development of a system that automates solar panel cleaning allowing for minimal human intervention. Experiments have validated its capabilities to effectively clean the solar panels with a 30 degree inclination, effectively increasing the energy output.

The automated drone based system as described in the paper by **Abhinav et al.** [17] allows the remote maintenance of solar panels. This increases the energy output of the system by removing dust and debris from the panel itself. The evaluation that is presented in the paper looks at the design, operational methods and performance. Two significant technological issues and autonomy issues as well as two operational challenges are explored in depth. The research works to create a cost-effective drone that tackles these issues and is cost effective to maintain the PV panel efficiency and enhance the energy output of the panels.

Most current automated solar panel cleaning technologies are based on a combination of scheduled cleaning and dry cleaning techniques; real-time performance-based decision-making is generally not used in operation. There has also been little emphasis on the integration of energy-efficient operation with wet cleaning to remove adhered dust. Thus,

creating an opportunity for a low-cost condition-based system that maximizes energy consumption as well as cleaning efficiency. Some studies also examined a predictive cleaning strategy driven by AI (artificial intelligence) as well as incorporating machine learning models and weather forecasts to predict dirt accumulation and schedule cleaning cycles based on those predictions. While these techniques would improve predicted dirt accumulation and reduce the need for scheduled cleaning; however, due to the need for cloud connectivity, large datasets, and many computing resources, the proposed system takes a different approach with a more lightweight threshold based design that is more appropriate for lower cost stand-alone photovoltaic systems.

III. METHODOLOGY

The proposed system is an autonomous cleaning system for solar panels that detects when the efficiency has dropped due to dirt and uses sensor inputs to determine whether or not conditions are suitable for performing a clean. It continuously monitors the power efficiency of the solar panel and if it senses the performance drop in normal brightness, then a cleaning cycle of the solar panel is initiated.

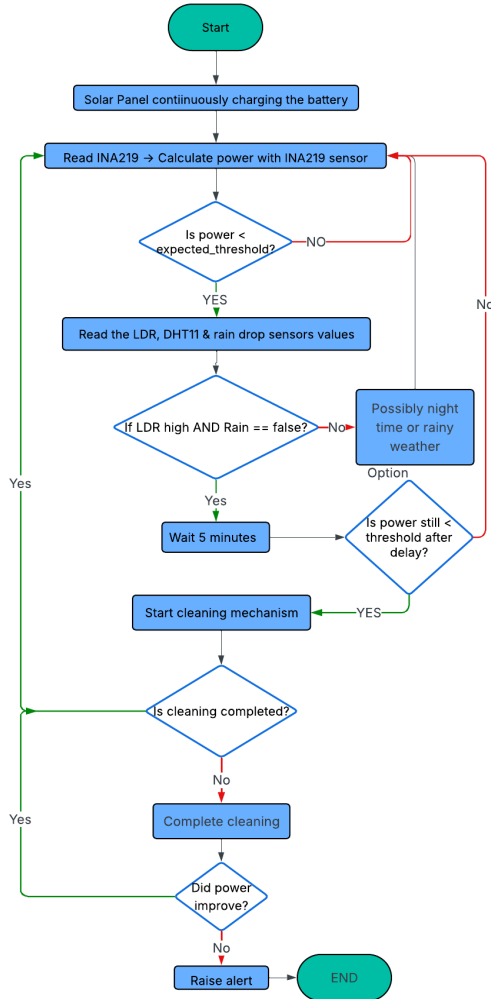


Fig. 1: Proposed system workflow for condition-based solar panel cleaning

A. System Architecture

An integrated system to monitor solar energy and clean solar panels automatically is proposed. The complete system workflow is shown in Fig. 1. This system includes four subsystems:

1. **Solar Panel Unit:** The panel is connected to the battery and the electronics system via the solar panel controller which simultaneously charges the battery and operates the load connected. The solar charge controller helps in regulating the incoming voltage from the solar panel and supplies it to the further connections.
2. **Energy Storage (Battery):** This can be continuously charged by the panel until it reaches the maximum capacity and later on it is cut off by the solar charge controller.
3. **Monitoring and Sensing Unit:** Various electronics sensors and the actuators are connected to the system for monitoring the environmental conditions and take actions as per the situation. It is further discussed in detail in the electronics section.
4. **Automated Cleaning Unit:** The cleaning mechanism includes a water pump which is initialized when the inefficiency is detected. Along with the water pump the wiper movement is also started which begins to clean the panel. Once the cleaning is done the water pump is turned off and the wiper returns to its original position.

B. System Design

In order to maintain clean solar panels, an automatic linear motor-driven cleaning assembly was proposed to be mounted directly onto the solar panel frame. Each unit of the cleaning system would be able to travel across the surface of a solar panel and remove dirt and debris that accumulates over time. The complete solar panel cleaning mechanism design can be seen in Fig. 2. The actual experimental setup is also shown in Fig. 3. The complete cleaning mechanism was designed using Autodesk Fusion and the 3D printed using the PLA printing filament. Once all the parts were printed they were all assembled together for a complete cleaning mechanism. The system was constructed to:

1. Create a stable mounting system
2. Provide smooth linear movement across the panels horizontally
3. Apply equal cleaning force across the pool of panels

4. Be structurally sound without stressing the panel itself

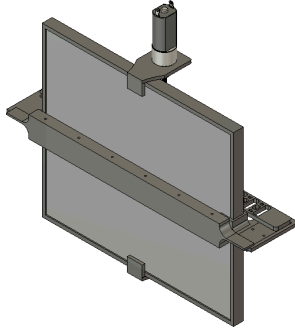


Fig. 2: (a) Panel cleaning mechanism front view

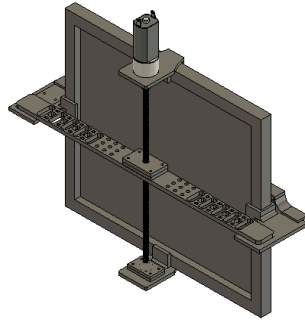


Fig. 2: (b) Cleaning mechanism back view

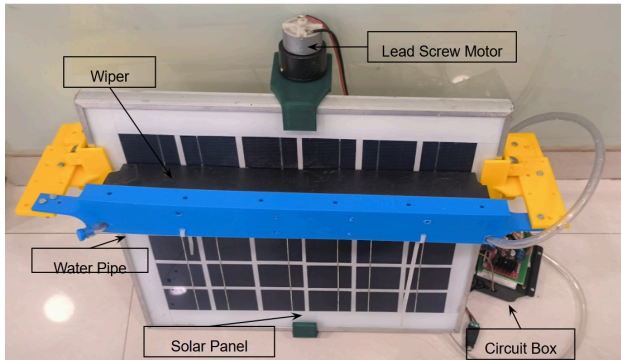


Fig. 3: Complete Experimental Setup

C. Electronics Requirements

The complete monitoring system is based on the ESP32 microcontroller. The ESP32 is connected to a INA219 current and voltage sensor which is further connected in series to the solar panel. This sensor was used to measure the current and the voltage generated by the solar panel which was further converted into the power efficiency rate for setting the cleaning thresholds. This value is continuously monitored by the microcontroller. Along with that a Light Dependent Resistor (LDR) is also connected which is used to detect the sunlight intensity and accordingly operate the cleaning mechanism. A DHT11 Temperature and humidity sensor with

a rain drop sensor is connected to the circuit for measuring the weather conditions and then in case of too cold or rainy weather the efficiency drop is justified. All the readings are visible on a 0.96 inch OLED display. All these sensors help in monitoring the environmental conditions as well as the solar panel output.

Through its correlation of the output of the panel with ambient light intensity and temperature readings, the system distinguishes between temporary efficiency loss due to a rise in air temperature and a permanent loss of efficiency due to the accumulation of dust. If there is a loss of output power that occurs at the same time that the temperature is elevated or there is low irradiance where the panels are generating energy, the cleaning will not be initiated. The cleaning will be initiated only after the reduction in output is under normal irradiance and within acceptable environmental conditions of temperature and wind indicating that a build-up of dust most likely has occurred.

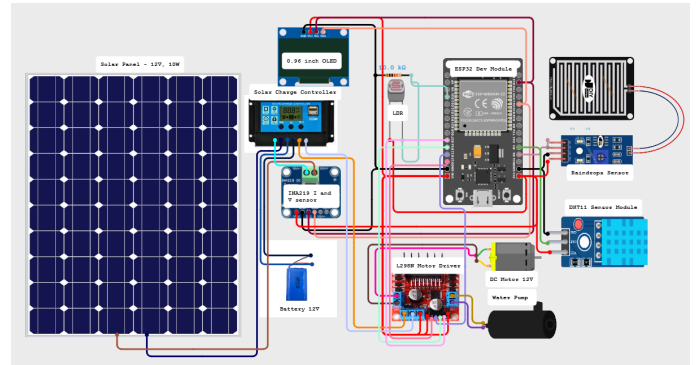


Fig. 4: Circuit Schematic of the complete system

The ESP32 microcontroller is connected with an L298N motor driver which is used to control the 12V 100 RPM DC Motor used for the rotary movement of the lead screw and linear movement of the wiper and the water pipe. The DC motor and the water pump are connected to the outputs of the motor driver and controlled as per the signal from the ESP32.

The solar panel used was a 12V and a 10W panel with a 36 cell arrangement. For regulating the signal coming from the solar panel a solar charge controller was used with a 10A rating. The circuit schematic of the complete system is given in Fig. 4. Similarly Fig. 5 shows the complete electronics circuit.

Long-term cleaning triggers may be impacted by both sensor drift and inaccuracies during calibration, particularly when working outside. Recalibrating both the INA219 sensor and the environmental sensors on a periodic basis will help to ensure that the threshold for threshold accuracy is maintained and to provide dependable long-term performance.

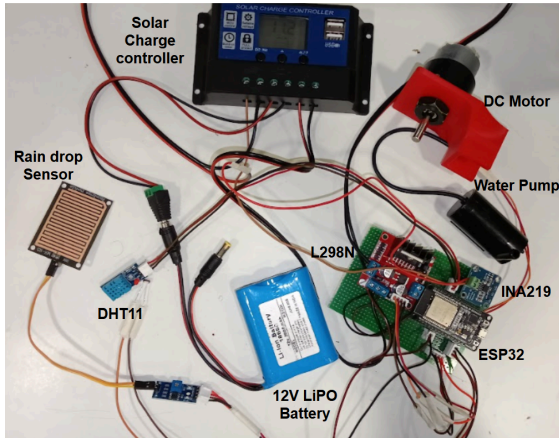


Fig. 5: Actual Components used

D. Solar Panel Performance Calculations

To better understand how dust affects PV performance, a detailed analytical model of the efficiency of panels and energy generated by the PV Panels was developed. The electrical and geometrical performance of the photovoltaic panel used in this study was estimated using the rated specifications of the module and the standard irradiance condition. The parameters used for the analytical evaluation of the photovoltaic panel of 10W are mentioned in Table I

Table I: Rated and Assumed Parameters of the Solar panel

Parameter	Symbol	Value	Unit
Maximum voltage	V_m	19.25	V
Maximum current	I_m	0.25	A
Rated output power	P_m	10	W
Length of one solar cell	L	0.045	m
Width of one solar cell	W	0.035	m
Number of cells	N	36	-
Standard irradiance	G	1000	W/m^2
Peak sun hours	t	5	h/day
Assumed panel efficiency under dusty condition	-	0.60	-

This calculation did not take into account any specific tilt angle or directional optimization for this analysis, which was based only on standard irradiance conditions. Therefore, any change in panel inclination would alter the duration and availability of effective solar exposure, however, the power

usage for cleaning and monitoring systems will remain constant.

Area of One Solar Cell: The area of one solar cell is calculated as -

$$A_{cell} = L \times W$$

(1)

Substituting the given values as shown in Eq. (1):

$$A_{cell} = 0.045 \times 0.035 = 0.001575 m^2$$

Total Area of the Solar Panel: The total effective panel area is the sum of the areas of all cells as per Eq. (2) -

$$A_{panel} = A_{cell} \times N$$

$$A_{panel} = 0.001575 \times 36 = 0.0567 m^2$$

Input Solar Power Incident on the Panel: The input solar power received by the panel under standard irradiance as per Eq. (3) is -

$$P_{in} = G \times A_{panel}$$

$$P_{in} = 1000 \times 0.0567 = 56.7W$$

Thus, under standard test irradiance, the total solar power incident on the panel surface is 56.7 W.

Output Electrical Power of the Solar Panel: The rated electrical output power of the panel is calculated as per Eq. (4) -

$$P_{out} = P_m = V_m \times I_m$$

$$P_m = 19.25 \times 0.25 = 4.8125W$$

However, the panel is specified as a 10 W rated module. Therefore, for practical and design-based analysis, the manufacturer-rated panel output is taken as -

$$P_{out} = 10 Watts$$

In order for the manufacturer of these devices to create an accurate representation calculation for us to perform based upon their products, the manufacturer uses the 10 watt rating for all future calculations for all future usage.

Efficiency of the Solar Panel: The efficiency of the panel is defined as per Eq. (5) as -

$$\eta = \frac{P_{out}}{P_{in}} \times 100$$

(5)

$$\eta = \frac{10}{56.7} \times 100 = 17.64\%$$

Therefore, the clean-condition efficiency of the solar panel is 17.64%

Effective Output Under Dusty Condition: To estimate the reduction in useful output due to dust deposition, it was assumed that the panel retains only 60% of its clean-condition performance. Thus, the effective output under dusty condition as per Eq. (6) is:

$$P_{dust} = 0.60 \times P_{out}$$

$$(6) \quad P_{dust} = 0.60 \times 10 = 6.0W$$

So, the solar panel output in the presence of dust is estimated as 6.0W as per Eq. (6). The model assumes that the dust is evenly distributed and does not account for partial shading or any uneven deposition of the dust on the panels, and these factors can also affect the actual performance of the panels. Similarly, the effective efficiency as per Eq. (5) under dusty condition becomes -

$$\eta_{dust} = 0.60 \times 17.64 = 10.58\%$$

Power Loss Due to Dust: The loss in output power due to dust accumulation is calculated as per Eq. (7) as-

$$P_{loss} = P_{clean} - P_{dust}$$

$$(7) \quad P_{loss} = 10 - 6 = 4W$$

Hence, the estimated power loss due to dust deposition is 4W.

Daily Energy Generation Estimate: Assuming an average of 5 peak sun hours per day, the daily electrical energy generated by the clean panel as per Eq. (8) is -

$$E_{clean} = P_{clean} \times t$$

$$E_{clean} = 10 \times 5 = 50Wh/day$$

$$(8)$$

For the dusty panel as per Eq. (9) -

$$E_{dust} = P_{dust} \times t$$

$$E_{dust} = 6 \times 5 = 30 Wh/day$$

$$(9)$$

Therefore, the daily energy loss due to dust as per Eq. (10) is:

$$E_{loss} = P_{loss} \times t$$

$$E_{loss} = 50 - 30 = 20Wh/day$$

$$(10)$$

By scaling the solar array over 30 W per panel, with a fixed cleaning schedule of once daily, and maintaining the net effectiveness of the entire self-sustaining unit(s), the overall benefit to the cleaning mechanism is evident through the ability to have higher scalability than that of the current setup. This can be seen in Table II.

Table II: Solar panel scaling to 30W

Parameter	10 W Panel	30 W Panel
Area (A_{panel})	0.0567 m ²	0.171 m ²
Input Power (P_{in})	56.7 W	170.1 W
Clean Efficiency (η)	17.64%	17.64%

Dusty Output	6 W	18 W
Daily Clean Energy (5 hrs)	50 Wh	150 Wh
Daily Dust Loss	20 Wh	60 Wh

Calculations show that the addition of the dust accumulated on the PV Panels has an impact on both instantaneous power output and daily energy production. Therefore, there is a need for a system to help with keeping the panels free from dust.

The ESP32-L298N architecture being described in this proposal should be an appropriate solution for small to medium sized photovoltaic systems; however, larger scale commercial solar farms will require distributed control architecture, industrial grade drivers for the motor control, and an overall centralized monitoring system to ensure the reliability of long distance systems, synchronization and improved power efficiency. The L298N driver design also causes some switching losses because it employs a bipolar transistor design. For large deployments, the use of higher efficiency MOSFET designed drivers will be more appropriate than the L298N driver.

VI. RESULTS AND ANALYSIS

A. Solar Panel Performance Analysis

Three operating states were tested for the photovoltaic panel being clean, having accumulated dust, and having accumulated dust and then being cleaned. The output power and amount of energy produced were measured and compared to show how much dust had an effect on the system's output as well as how effective the cleaning method was at improving that output. This is demonstrated in the Fig. 6 graph which shows the before and after cleaning situation.

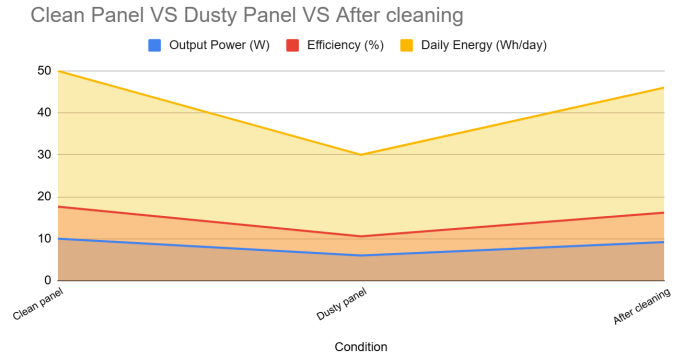


Fig. 6: Before and after Cleaning outcomes

B. Energy Loss vs System Consumption

To evaluate the ease of the proposed system, the daily energy loss due to dust was compared with the energy consumed by the monitoring and cleaning system. All these values are mentioned in Table III. The total energy wasted from our system from dust particle building up is 20 watt hours per day, while the total amount of energy used by our system is 19.5

watt hours per day. This shows that the system is working with a net positive energy balance, meaning it is viable from an energy standpoint. Although there is a small overall gain for this particular 10W prototype, it demonstrates that it is possible to make these systems functional. The majority of total energy usage is from monitoring electronics, while actual cleaning cycles use relatively little energy. Increased solar panel capacity will greatly decrease the amount of energy used for overhead, which will help to increase the practicality of using this system on a larger scale.

Table III: Energy losses and consumptions

Parameter	Value (Wh/day)
Clean panel energy	50
Dusty panel energy	30
Energy loss due to dust	20
System consumption	19.5
Net energy gain	0.5

C. Operational Energy Distribution

The total daily energy consumption was further analyzed based on operational modes to understand the contribution of idle and active states. These are as mentioned in Table IV.

Table IV: Operational energy consumption

Mode	Energy (Wh/day)
Idle monitoring	19.2
Cleaning cycle	0.3
Total	19.5

D. Scalability Analysis

To evaluate scalability, the system performance was analyzed for a higher capacity panel while keeping system consumption constant. The scalability graph is as shown in Fig. 7. The findings indicate a decrease in system's relative energy overhead according to increased panel capacity. This proves that the proposed system has excellent scalability and increased efficiency for larger solar projects.

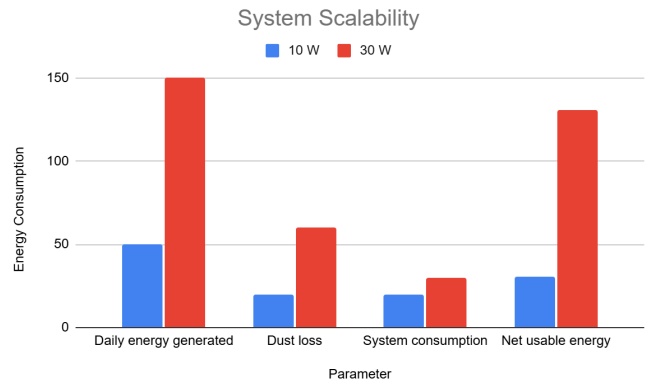


Fig. 7: Scalability analysis graph

E. Comparative Analysis with Existing Methods

The proposed system outperforms conventional methods by achieving ~50% efficiency improvement under dusty conditions while reducing unnecessary energy and water usage, making it more efficient and cost-effective. This can be understood with the help of Table V.

Table V: Comparison of Proposed System with Existing Cleaning Methods

Method	Cost (Implementation & Maintenance)	Efficiency Improvement	Energy Consumption	Limitations
Manual cleaning	Low initial cost, high recurring labor cost	30–40%	Human dependent	Labor-intensive, inconsistent, water wastage
Dry robotic cleaning	Medium to high	20–35%	Medium	Ineffective for sticky/wet dust, complex design
Scheduled automated cleaning	Medium	25–40%	Medium-high	Cleans unnecessarily, energy wastage
Proposed system	Low to medium, minimal maintenance	~50% improvement (dusty → cleaned)	Low (19.5 Wh/day)	Dependent on sensor calibration, L298N inefficiency

The analytical and experimental data back up the proposed system's ability to minimize power losses from dust, while at the same time minimising total energy consumption. Because total system consumption had been continuously measured, the total system still had lower energy consumption than the energy loss due to dust build-up. Thus, providing greater efficiency and energy production for the photovoltaic system and substantiating its practical application. The analysis assumes uniform dust distribution and constant irradiance. Real-world variations may introduce deviations in measured performance.

According to the research findings, using condition-based cleaning increases productivity and reduces unused actuators and water. The result shows that it is possible to create lightweight embedded monitoring systems for autonomous maintenance of photovoltaic panels.

VII. CONCLUSION

The proposed solar panel cleaning mechanism presents a new, energy-efficient method to remove dust based on conditions of operation that are responsible for performance degradation caused by accumulated dust. Experiments and analysis showed that dust can reduce the output power of a solar panel by up to 40%. The cleaning of the panels using the system brings the panel output back to approximately 90% to 92% of the output before it was dusty; thus, the use of the cleaning system represents a nearly 50% increase in output compared to when the panels are covered with dust. In addition, when considered over an extended period of time, the system also provides a net positive energy balance for each panel because the daily amount of energy lost to the dust (20 W/day) is greater than the total energy consumed by the system (19.5 W/day). This shows that the system is both energy neutral and cost-efficient for small solar photovoltaic installations. Future developments will include adding maximum power point tracking (MPPT), developing a more efficient motor driver, developing adaptive cleaning algorithms, adding internet of things (IoT) monitoring capabilities, and improving the mechanical designs.

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